# RISE Internship
# Machine Learning & AI
# Project List

**Tools:**

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

## Project 1: Industry-Oriented Machine Learning System for Customer Churn Prediction

**Problem Statement:**
Businesses lose significant revenue due to customer churn but often fail to identify customers who are likely to leave. Manual analysis is ineffective for large datasets. Industry uses machine learning models to predict churn and take proactive retention actions.

**Objective:**
To build a machine learning model that predicts customer churn using historical customer behavior and transaction data.

**Requirements:**

- Customer dataset ingestion
- Data cleaning and preprocessing
- Feature selection and engineering
- Exploratory data analysis
- Machine learning model training

- Model evaluation using accuracy and recall metrics
- Prediction of high-risk customers
- Visualization of churn factors
- Model documentation

## Expected Outcome:

- Hands-on experience with supervised machine learning
- Understanding of business-driven ML use cases
- Industry-ready machine learning portfolio project
- Job relevance for Machine Learning Engineer, Data Scientist, Business Analyst roles

**Tools:**

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

## Project 2: Industry-Oriented Machine Learning System for Credit Risk Assessment

**Problem Statement:**

Banks and financial institutions face high risk when approving loans without accurate risk evaluation. Manual credit assessment is slow and subjective. Industry uses machine learning models to assess credit risk and reduce default rates.

**Objective:**

To build a machine learning model that evaluates credit risk and predicts loan default probability.

**Requirements:**

- Financial dataset ingestion
- Data preprocessing and normalization
- Feature engineering for credit indicators
- Exploratory data analysis
- Model training using classification algorithms

- Model evaluation using precision and recall
- Risk score generation
- Visualization of key risk factors
- Model documentation

**Expected Outcome:**

- Hands-on experience with financial ML applications
- Understanding of risk-based modeling
- Industry-ready machine learning portfolio project
- Job relevance for Machine Learning Engineer, Risk Analyst, Data Scientist roles

**Tools:**

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

## Project 3: Industry-Oriented Machine Learning System for Fraud Detection

**Problem Statement:**
Financial institutions and e-commerce platforms face increasing fraudulent transactions that result in major financial losses. Rule-based systems fail to detect evolving fraud patterns. Industry relies on machine learning models to identify suspicious activities in real time.

**Objective:**
To build a machine learning model that detects fraudulent transactions based on transaction behavior and historical data.

**Requirements:**

- Transaction dataset ingestion
- Data cleaning and preprocessing
- Handling class imbalance
- Feature engineering for fraud indicators
- Machine learning model training

- Model evaluation using precision, recall, and F1-score
- Visualization of fraud patterns
- Prediction of suspicious transactions
- Model documentation

**Expected Outcome:**

- Hands-on experience with anomaly and fraud detection
- Understanding of imbalanced dataset handling
- Industry-ready machine learning portfolio project
- Job relevance for Machine Learning Engineer, Fraud Analyst, Data Scientist roles

**Tools:**

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

## Project 4: Industry-Oriented Machine Learning System for Employee Attrition Prediction

**Problem Statement:**
Organizations face high costs due to employee attrition and struggle to identify factors that cause employees to leave. Manual HR analysis is ineffective for large workforces. Industry uses machine learning to predict attrition and improve retention strategies.

**Objective:**
To build a machine learning model that predicts employee attrition based on HR and performance data.

**Requirements:**

- HR dataset ingestion
- Data cleaning and preprocessing
- Feature engineering for attrition factors
- Exploratory data analysis
- Machine learning model training

- Model evaluation using accuracy and recall
- Identification of key attrition drivers
- Visualization of employee trends
- Model documentation

**Expected Outcome:**

- Hands-on experience with HR analytics use cases
- Understanding of workforce-related ML problems
- Industry-ready machine learning portfolio project
- Job relevance for Machine Learning Engineer, HR Analyst, Data Scientist roles

**Tools:**

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

## Project 5: Industry-Oriented Machine Learning System for Sales Forecasting

**Problem Statement:**

Businesses struggle to predict future sales accurately, leading to poor inventory planning and revenue loss. Traditional forecasting methods fail to capture complex trends and seasonality. Industry relies on machine learning models for accurate sales forecasting.

**Objective:**

To build a machine learning model that predicts future sales using historical sales and business data.

**Requirements:**

- Historical sales dataset ingestion
- Data preprocessing and feature engineering
- Trend and seasonality analysis
- Machine learning model training
- Model evaluation using regression metrics

- Forecast visualization
- Business insight generation
- Model documentation

**Expected Outcome:**

- Hands-on experience with regression and forecasting models
- Understanding of time-based ML problems
- Industry-ready machine learning portfolio project
- Job relevance for Machine Learning Engineer, Data Scientist, Business Analyst roles

**Tools:**

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

## Project 6: Industry-Oriented Machine Learning System for Recommendation Engines

**Problem Statement:**
E-commerce, OTT, and EdTech platforms struggle to personalize content for users at scale. Rule-based recommendation systems fail to adapt to user behavior. Industry uses machine learning models to deliver personalized recommendations that improve engagement and conversions.

**Objective:**
To build a machine learning–based recommendation system that suggests products, movies, or courses based on user behavior and preferences.

**Requirements:**

- User interaction dataset ingestion
- Data preprocessing and feature encoding
- User and item similarity analysis
- Model training using recommendation algorithms

- Evaluation using recommendation metrics
- Personalized recommendation generation
- Visualization of recommendation patterns
- Model documentation

**Expected Outcome:**

- Hands-on experience with recommendation systems
- Understanding of personalization techniques
- Industry-ready machine learning portfolio project
- Job relevance for Machine Learning Engineer, Data Scientist, Recommendation Systems Engineer roles

**Tools:**

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

## Project 7: Industry-Oriented Machine Learning System for Image Classification

**Problem Statement:**
Industries such as manufacturing, healthcare, and retail rely on image-based systems for quality inspection, diagnosis, and automation. Manual inspection is slow and error-prone. Industry uses machine learning and computer vision models to classify images accurately.

**Objective:**
To build a machine learning model that classifies images into predefined categories using image features and ML algorithms.

**Requirements:**

- Image dataset selection
- Image preprocessing and normalization
- Feature extraction from images
- Model training using classification algorithms
- Model evaluation using accuracy metrics

- Visualization of classification results
- Testing on unseen images
- Model documentation

## Expected Outcome:

- Hands-on experience with image-based ML systems
- Understanding of computer vision fundamentals
- Industry-ready machine learning portfolio project
- Job relevance for Machine Learning Engineer, Computer Vision Engineer, AI Engineer roles

**Tools:**

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

## Project 8: Industry-Ready End-to-End Machine Learning System for Business Intelligence

**Problem Statement:**
Many organizations experiment with machine learning models but fail to integrate them into end-to-end business workflows. Isolated models without insights, evaluation, and reporting do not create business value. Industry requires complete ML systems that support decision-making.

**Objective:**
To design and implement an end-to-end machine learning solution that transforms raw business data into predictions, insights, and actionable intelligence.

**Requirements:**

- Business problem definition
- Data collection and preprocessing pipeline
- Exploratory data analysis
- Feature engineering
- Machine learning model selection and training

- Model evaluation and comparison
- Prediction and insight generation
- Visualization of business outcomes
- Complete project documentation

**Expected Outcome:**

- End-to-end experience with real-world ML pipelines
- Strong industry-grade machine learning capstone project
- Understanding of how ML supports business decisions
- Job relevance for Machine Learning Engineer, Data Scientist, AI Analyst roles